

MARY LETEY

maryletey@fas.harvard.edu

EDUCATION

Sept 2023 – Present	Harvard University	PhD – Applied Mathematics
Sept 2022 – June 2023	Perimeter Institute	MSc – Theoretical Physics
Sept 2018 – June 2022	University of Cambridge	BA Honours – Mathematics

PUBLICATIONS & PREPRINTS

Sequential correlations change in-context learning: Effective context length and architectural mismatch
[HiLD Workshop at ICML 2026](#) • **M. Letey**, Lu, Pehlevan, Zavatone-Veth

Pretrain–test task alignment governs generalization in in-context learning
[ICLR 2026](#) • **M. Letey**, Zavatone-Veth, Lu, Pehlevan

Theory of scaling laws for in-context regression: Depth, width, context and time
[ICLR 2026](#) • Bordelon, **M. Letey**, Pehlevan

Asymptotic theory of in-context learning by linear attention
[Proceedings of the National Academy of Sciences \(2025\)](#) • Lu, **M. Letey**, Zavatone-Veth, Maiti, Pehlevan

Quantum initial conditions for curved inflating universes
[Physical Review D \(2024\)](#) • **M. Letey**^{*}, Shumaylov^{*}, Agocs, Handley, Hobson, Lasenby

INVITED TALKS

EPFL – [FLARE Workshop](#), May 2026
Invited to present on theoretical models of in-context learning

SIAM – Conference on Uncertainty Quantification, March 2026
Invited to present “Asymptotic theory of in-context learning by linear attention”

Kempner Institute – [Spring Into Science Retreat](#), May 2024
Opening talk on “Asymptotic theory of in-context learning by linear attention”

WORK & RESEARCH EXPERIENCE

Research Internship Winter 2026	Hudson River Trading Machine learning for market prediction
Research Internship Summer 2023	Quebec Artificial Intelligence Institute (MILA) Equivariant architectures for neural PDE solvers
Master’s thesis Sept 2022 – Sept 2023	Perimeter Institute for Theoretical Physics Classification of discrete symmetries in non-Euclidean space
Research Internship Summer 2022	Cavendish Laboratory & Kavli Institute for Cosmology, Cambridge Research on early-universe cosmology and curved spacetimes
Internship Summer 2020	Embryo Ventures Due diligence and founder outreach
Research Internship Fall 2019 – Spring 2020	Iguana Investments Market prediction models

AWARDS & FUNDING

2025	Graduate Fellowship	Kempner Institute – Full support for graduate studies
2023	Graduate Prize Fellowship	Harvard University
2022	International Scholar Award	Perimeter Institute – Full support for MSc

TEACHING

2025	Harvard University	TF for APMATH 50
21-23	Blue Education	Maths, physics, and university admissions tutoring